Epidemiology basics

Definition, the epidemiological triad, natural history of disease and measures of frequency — with examples from dentistry

Practical Class 8 · Dental Public Health · 2026/2027

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What you will be able to do after this class

1. Define epidemiology and describe a problem by **person, place and time**.
2. Apply the **epidemiological triad** and the **natural history of disease** to caries and periodontitis.
3. Distinguish **proportion, rate and ratio**.
4. Calculate **prevalence**, **cumulative incidence**, **incidence rate** (with person-time), **mortality** and **case-fatality**.
5. Explain why **crude** rates can mislead and what standardisation does.
6. Name the main **data sources** in Bulgaria.

Plan of the class (90 min)

1. What epidemiology is — 14 min
2. Triad and natural history — 10 min
3. Measures of frequency — 41 min
4. Crude and specific rates — 8 min
5. Data sources — 6 min
6. Exercises and recap — 11 min

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1. **What epidemiology is**
2. Epidemiological triad and natural history of disease
3. Measures of disease frequency
4. Crude and specific rates
5. Sources of epidemiological data
6. Exercises

Definition: six key words

> *Epidemiology is the study of the **frequency, distribution and determinants** of diseases and other health-related conditions in **human populations**, and the **application** of this study to the promotion of health and to the prevention and control of health problems.*
>
> — Kebede (2004), after Last, A Dictionary of Epidemiology

Key word	**Meaning and dental example**
population	the unit is the group, not the individual: all 12-year-olds in the region, not the child in the chair
frequency	a quantitative science: how many people, how many new cases, over what time
distribution	**person, place, time**: e.g. 12-year-olds in villages around Plovdiv have higher DMFT than in the city (Onov & Beltcheva, 2020)
determinants	**risk** factors (sugars, tobacco) and **protective** factors (fluoride, sealants)
health-related conditions	not only diseases: injuries, behaviour (toothbrushing), access to dental care
application	the aim is action: prevention programmes, service planning, evaluation of effect

From the patient in the chair to the health of the community

Clinical question

Does this child have untreated caries? Examine the teeth, make a diagnosis and develop an individual care plan.

The decision concerns the **patient**.

Epidemiological question

How many children in this school have untreated caries? Examine a defined group using the same criteria and calculate the frequency.

The decision concerns the **population**.

Teaching example

Of 200 examined children, 80 have untreated caries: **80 / 200 = 40%**. This helps plan care for the group; it does not mean that each child has "40% caries".

Teaching example; definition and uses of epidemiology — Kebede (2004), unit 1

What epidemiology is used for

Community diagnosis

What are the oral health problems of the population and how common are they? (e.g. the share of 12-year-olds with untreated caries)

Finding causes

Which factors raise the risk? Sugar and caries, tobacco and periodontitis, HPV and oropharyngeal cancer.

Natural history and prognosis

How does disease progress without intervention? How fast does an initial lesion become a cavity?

Evaluating interventions

Does fluoride varnish reduce caries? Is a sealant programme worth it?

Planning services

How many dentists, children's clinics and prosthetic services does an ageing population need?

Surveillance

Ongoing collection, analysis and dissemination of data **for action** — e.g. oral cancer incidence by year.

Kebede Y. Epidemiology, University of Gondar (2004), unit 1; Burt & Eklund, 7e (2021), ch. 12

A short history: from observation to study

Year	Event
5th c. BC	Hippocrates, "Airs, Waters, Places" — environment and way of life influence health
1662	John Graunt analyses the London "Bills of Mortality" — the beginning of health statistics
1747	James Lind compares six treatments for scurvy in sailors — an early systematic comparison
1854	John Snow maps cholera around the Broad Street pump — the "father of epidemiology"
1948–1954	Framingham: cardiovascular risk factors; Doll and Hill: smoking and lung cancer

The dental story: fluoride

- **1901–1916**: Frederick McKay describes the "Colorado brown stain" — later recognised as **fluorosis**, in which caries is **less common**.
- **1939–1942**: Dean et al. — studies in **21 cities** (8 + 13): a historical finding: as water fluoride rises to about 1 mg/l caries falls, while fluorosis increases with concentration.
- **1945**: Grand Rapids, USA — the first controlled fluoridation of drinking water.

Burt & Eklund, ch. 12, 19, 24, 25 · Dean HT et al. Public Health Rep. 1942;57:1155. doi:10.2307/4584182 · Doll & Hill. BMJ. 1950;2:739. doi:10.1136/bmj.2.4682.739 · Dawber TR et al. AJPH. 1951;41:279. doi:10.2105/AJPH.41.3.279

Describe a health problem by "who, where, when"

Who? — person

Age, sex, social circumstances, dentition.

Example: 12-year-old children with permanent teeth.

Where? — place

Town, village, school, region; compare clearly defined populations.

Example: urban and rural areas in one region.

When? — time

Examination date, year, trend.

Example: change between two examinations using the same criteria.

From description to a question

"In this age group, untreated caries is more common in the villages." This raises questions about access, diet and prevention; the difference itself **does not establish its cause**.

CDC, Introduction to Epidemiology, Public Health 101; teaching examples

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The epidemiological triad: host, agent, environment

Element	In dental caries
host	tooth (morphology, enamel), saliva (flow, buffering), age, immunity
agent	dysbiotic dental biofilm with acid-producing bacteria (e.g. **S. mutans** — important, but neither the only nor a sufficient cause)
environment	frequency and amount of sugars, fluoride, hygiene habits, income, access to care
time	duration of exposure — caries is a slow process

Keyes' triad (historical model)

Caries develops where **tooth**, **microflora** and **substrate** (sugars) intersect. Today caries is seen as a dynamic, biofilm-mediated disease strongly driven by diet.

Causal pie model

A sufficient cause is a **complex** of components. Sugar is a **component** cause: not everyone who eats sugar develops caries, but removing it blocks every causal complex that contains it.

Burt & Eklund, 7e (2021), ch. 12: epidemiological triad, Keyes, Rothman

Natural history of disease and levels of prevention

Stage	In dental caries	Level of prevention
1. susceptibility	no disease, but risk factors present: frequent sugars, little fluoride, plaque	**primary**: health education, fluoride, sealants, less sugar
2. pre-symptomatic (subclinical)	early non-cavitated ("white spot") lesion — painless but clinically detectable; it can **arrest or remineralise**	**secondary**: early detection (check-ups, screening), remineralisation
3. clinical	cavitation, pain, pulpitis	**secondary/tertiary**: restoration, endodontic treatment
4. outcome	without treatment: arrest of the lesion or progression to abscess and tooth loss	**tertiary**: prosthetics, rehabilitation

Early management of initial caries can avoid more invasive treatment. Absence of pain does not mean absence of disease.

Kebede Y. Epidemiology (2004), unit 3; DPH/10-epidemiology.qmd

The level of prevention depends on the stage and the aim

Condition	Aim of action	Example
Sound tooth surface	**Primary**: prevent disease onset	Sugar control; fluoride prevention
Early lesion without pain	**Secondary**: detect and control disease early	Detection at examination; non-operative lesion management
Lost teeth and impaired chewing	**Tertiary**: limit the consequences	Prosthetic replacement and rehabilitation

Absence of symptoms

An initial carious lesion can be **visible at examination** while causing no pain.

One measure — different aims

Fluoride can protect a sound surface or help control an existing early lesion. The context determines the level.

Teaching applications of natural history; Kebede (2004), unit 3; Burt & Eklund (2021)

The iceberg of disease

Registered cases are the **visible tip**. Below the water remain:

- asymptomatic and early cases;
- people who do not seek care (fear, cost, distance);
- self-treatment;
- inaccurate diagnosis or recording.

In oral diseases the iceberg is large

Periodontitis and early caries do not hurt for a long time; some people see a dentist **only when in pain**. Practice reports show the people **served**, not all the people **affected**; prevalence among attenders may even be higher than in the population (selection).

Consequence

Estimating population prevalence requires **active examinations** of a representative sample — a national oral health survey following the WHO methodology (next class).

Department textbook of social medicine (BG), ch. 15: the iceberg of morbidity

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Why the number of cases is not enough

Example

In one year town A registers **30** new cases of oral cancer and town B **15**. Where is incidence higher?

Town	New cases	Population	Incidence (annual)
A	30	600,000	30 / 600,000 × 100,000 = **5 per 100,000**
B	15	150,000	15 / 150,000 × 100,000 = **10 per 100,000**

The absolute number

tells **how much** resource is needed: examinations, treatments, beds.

The rate

shows the **frequency** of new cases per unit of time — it allows comparison between groups of different size.

The multiplier 10^n (100, 1000, 100,000) is chosen to give a convenient number. Before counting — a **case definition**: diagnostic threshold, dentition, unit (person, tooth, surface).

Before counting, define a “case”

Working definition for this teaching example

A case is a **12-year-old child with at least one permanent tooth with untreated caries**, detected at examination using pre-specified diagnostic criteria.

Specification	Why is it needed?
Who?	12-year-old children; primary and permanent teeth are kept separate
What?	Untreated caries at a uniform threshold; “I have had toothache” is a different measure
When?	On the examination date; this is not the number of new cases in a year
Which unit?	A child with at least one affected tooth; three decayed teeth still give **one case**

Changing the definition or diagnostic threshold can change the number even in the same group.

Teaching definition; consistent measurement — WHO, Oral Health Surveys (2013)

“Percentage of children” and “percentage of teeth” differ

One group, two units

20 children × 20 primary teeth = **400 examined teeth**.

10 children each have 4 teeth with untreated caries; the other 10 have none. Total: **40 affected teeth**.

Count children

10 affected / 20 examined

50% Percentage of children with at least one tooth with untreated caries.

Count teeth

40 affected / 400 examined

10% Percentage of examined primary teeth with untreated caries.

The numerator and denominator must use the same unit: people with people, teeth with teeth.

Teaching data and calculations; people meeting a case definition differ from affected teeth

Choose the denominator to match the question

Question	Numerator	Denominator
How many children have caries **today**?	Children with caries at examination	All examined children
How many develop it **for the first time**?	Newly affected children during the period	Children without this caries at the start
How quickly do **new cases** occur?	New cases using the same definition	Time observed and at risk

The same school: 300 children

120 already have caries in permanent teeth; 180 have none.
Prevalence at the start: **120 / 300 = 40%**.
Later, 45 of the 180 develop their first caries: **45 / 180 = 25% over 3 years**.

Existing cases count towards prevalence but are outside the denominator for first disease occurrence.

Teaching example; CDC, Principles of Epidemiology, lesson 3, section 2

Proportion, rate, ratio

Measure	Definition	Formula	Dental example
proportion (share)	the numerator is **part of** the denominator; 0–1 or %	a / (a + b)	65 of 100 children have caries experience → 65%
rate	new events per unit of **time** at risk	events / total time at risk	12 new carious lesions per 100 person-years (one person may contribute several lesions)
ratio	one quantity divided by another; the numerator **need not** be part of the denominator (a proportion is a special case)	a / b	dentists per 10,000 population; men : women among cases

A ratio from Bulgaria

WHO estimates for 2020: 317 new cases of lip and oral cavity cancer in men and 138 in women → ratio **2.3 : 1**.

WHO. Oral Health Country Profile – Bulgaria (2022)

Prevalence

Prevalence — the proportion of people in a population who **have** the disease at a given time or during a given period (**new and existing** cases).

$$P = \frac{\text{people meeting the case definition}}{\text{population examined}} \times 10^n$$

- **point** prevalence — at a particular date;
- **period** prevalence — at any time during the interval, each person counted once.

A measurement at a specified date is a "snapshot" of the population.

Prevalence rises with	**Prevalence falls with**
longer duration of disease	high case-fatality
more effective supportive treatment (fewer deaths)	rapid recovery
higher incidence	lower incidence
better diagnosis	immigration of healthy people
immigration of cases, emigration of healthy people	emigration of cases

Department textbook of social medicine (BG), ch. 15

Six people over one year: three “snapshots”

Hypothetical illness allowing recovery; all 6 people are observed throughout the year.

Person	1 January	1 July	31 December	What happens?
A	ill	well	well	recovers in April
B	ill	ill	ill	ill throughout the year
C	well	ill	ill	new case in March
D	well	well	ill	new case in September
E	well	well	well	does not become ill
F	well	well	well	does not become ill

Point prevalence: January 2/6; July 2/6; December 3/6. Each time count the people ill on that date.

Entirely hypothetical data; 2/6 = 33.3%; 3/6 = 50%

Incidence (1): cumulative incidence — risk

Incidence — the frequency of **new** cases during a defined period in the **population at risk**.

$$\text{CI} = \frac{\text{new cases during the period}}{\text{people at risk at the start of the period}}$$

- a probability (0–1); always with a **period**: "12% over 3 years";
- the denominator **excludes** existing cases and those who cannot develop the disease;
- this simple calculation assumes a **fixed group**, initially at risk, and complete ascertainment of outcomes.

Example

In 2023 a school has 300 six-year-olds; 120 already have caries in the first permanent molars. The remaining **180** children without caries in permanent teeth are followed for 3 years; **45** develop caries.

$$\text{CI} = \frac{45}{180} = 0.25 = 25\% \text{ over } 3 \text{ years}$$

Dividing by 300 would be wrong: 120 children already meet the case definition and cannot become a **new** case (event = first caries in a permanent tooth of the person).

The same six people give different, valid results

Measure	Who is counted?	Calculation
Point prevalence on 31 December	B, C, D — ill on that date	3 / 6 = **50%**
Period prevalence for the year	A, B, C, D — ill at least once	4 / 6 = **66.7%**
Cumulative incidence for the year	C and D are new cases among C, D, E, F, who were initially well	2 / 4 = **50%**

Period prevalence is not a sum

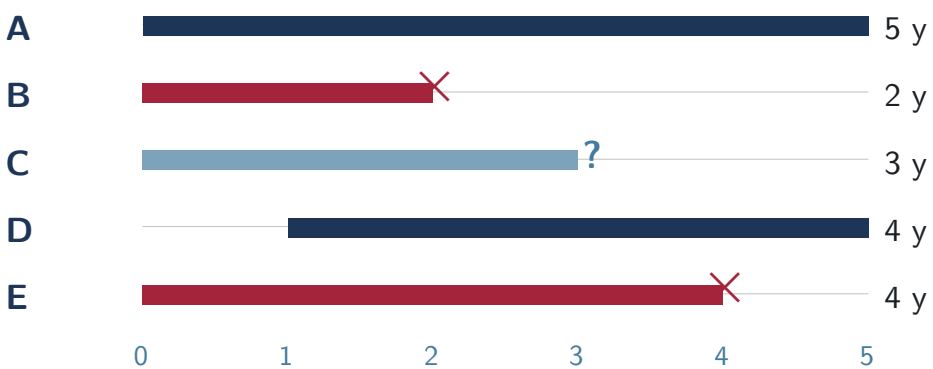
Do not add 33.3% + 33.3% + 50%. A person can appear in several snapshots but is counted **once** for the period.

A numerical coincidence

December prevalence and annual risk are both 50% here, but their **numerators and denominators differ**.

Calculations from the preceding six-person table; definitions from CDC, lesson 3

Incidence (2): incidence rate and person-time



× new case of periodontitis · ? lost to follow-up · years of follow-up 0–5

$$\text{IR} = \frac{\text{new cases}}{\text{total person-time at risk}}$$

Person-years	**5 + 2 + 3 + 4 + 4 = 18**
New cases	2 (B and E)
IR	2 / 18 = **11.1 per 100 PY**

The rate is **not a probability**: its unit is "per person-year" and it has no upper limit. Person-time handles unequal follow-up, but **informative drop-out** (e.g. sick people leaving more often) can still bias the result.

Person-time adds each person's observed interval at risk

Person	Observation within one calendar year	Time at risk
A	well and observed throughout the year	12 months
B	becomes ill at month 6; stops contributing time	6 months
C	well until lost to follow-up at month 9	9 months
D	enters for the last 3 months; remains well	3 months

First convert the time

$12 + 6 + 9 + 3 =$ **30 person-months**

$30 / 12 =$ **2.5 person-years**

Then divide events by time

1 new case / 2.5 PY = 0.4 per PY

= **40 new cases per 100 PY**

"40 per 100 person-years" is a rate of occurrence; it is not an observed annual risk of "40%".

Hypothetical exact intervals; CDC, Principles of Epidemiology, lesson 3, section 2

Prevalence, cumulative incidence, incidence rate

	Prevalence	**Cumulative incidence**	**Incidence rate**
numerator	all cases (new + existing)	new cases	new cases
denominator	whole population examined	people at risk at the start	person-time at risk
time	point or period	period must be stated	built into the unit
range	0–1	0–1 (probability)	0 to ∞, "per PY"
data needed	disease status on a date or during a period	initial status and new cases; complete follow-up	new cases and individual time at risk
answers	how many are ill now?	what is my risk of falling ill?	how fast do new cases appear?

For a rare event and a short period: $\mathrm{CI} \approx \mathrm{IR} \times t$. E.g. IR = 2 per 1000 PY over 5 years $\longrightarrow$ CI $\approx$ 1%.

Prevalence and incidence: $P \approx \mathrm{IR} \times D$

The bathtub analogy

The tap = incidence (new cases). **The water** = prevalence. **The drain** = recovery or death. The slower it drains (long disease), the more water accumulates.

Conditions: stable population, constant incidence and duration, rare disease; the time units of IR and D must match (in the simple equilibrium model for a more common disease: $\frac{P}{1-P} = \mathrm{IR} \times D$).

Example 1 · hypothetical stable population

IR of oral cancer = 3.3 per 100,000 PY; mean duration (from diagnosis to recovery or death) ≈ 4 years $\longrightarrow P \approx 3.3 \times 4 = 13.2$ per 100,000.

Example 2 · aphthous ulcer

Frequent but short episodes (about 10 days) give a **low** point prevalence.

Hypothetical example; 3.3 is close to the age-standardised incidence for Bulgaria (WHO, 2022), but a standardised rate is not used for this calculation

Higher prevalence does not prove that more new cases occur

Two teaching models: stable population, rare disease, constant incidence rate and mean duration.

Model	New cases	Mean duration	Approximate prevalence
A	2 per 1000 PY	1 year	$2 \times 1 =$ **2 per 1000**
B	2 per 1000 PY	5 years	$2 \times 5 =$ **10 per 1000**

What is the same?

The frequency of **new** cases. In model B people remain ill longer, so more existing cases accumulate.

What should we check?

Whether the difference reflects new cases, longer illness, survival, case detection or changes in the population.

To interpret a change in prevalence, also consider disease occurrence and duration.

Teaching arithmetic using $P \approx IR \times D$; these are not data from real populations

Mortality and case-fatality

Measure	Formula
crude death rate	deaths in the year / mid-year population × 1000
cause-specific	deaths from the cause / mid-year population × 10^n
age- (or sex-) specific	deaths in the group / mid-year population **of the same group** × 10^n
one-year case-fatality proportion	deaths from the disease within 12 months of diagnosis / patients in the same diagnosis group × 100
proportional mortality	deaths from the cause / all deaths × 100

Example (hypothetical town, 250,000 people, 2025)

- 60 new cases of oral cancer;
- each followed **12 months after diagnosis**: 18 died of the cancer;
- in 2025, 24 people in total died of oral cancer (including people diagnosed earlier).

incidence	**60/250,000 = 24 per 100,000**
cause-specific mortality	24/250,000 = **9.6 per 100,000**
one-year case-fatality	18/60 = **30%**

Mortality, proportionate mortality and case-fatality

The same hypothetical town: 250,000 people; 2400 total deaths in the year; 24 from oral cancer. Of 60 newly diagnosed patients, 18 die of the cancer within 12 months of diagnosis.

Measure	Calculation	How to read it
Cause-specific mortality	24 / 250,000 × 100,000 = **9.6 per 100,000**	Relative to the whole population, for the year
Proportionate mortality	24 / 2400 × 100 = **1%**	This cause's share of all deaths
One-year case-fatality	18 / 60 × 100 = **30%**	Share dying of the disease among the same diagnosed patients

Why is 24 / 60 not one-year case-fatality?

The 24 calendar-year deaths include people diagnosed earlier. They are not necessarily members of the 60 newly diagnosed cases.

Teaching extension of the previous example; CDC, Principles of Epidemiology, lesson 3, section 3

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Crude prevalence can mislead: edentulism in two towns

	Town A: population	Town A: edentulous	Town B: population	Town B: edentulous
20–64	8,000	160 (**2%**)	5,000	100 (**2%**)
65+	2,000	400 (**20%**)	5,000	1,000 (**20%**)
total (crude)	10,000	560 $\longrightarrow$ **5.6%**	10,000	1,100 $\longrightarrow$ **11.0%**

Age-specific prevalence

Calculated for a subgroup (age, sex). Here it is **identical** in the two towns.

Crude prevalence

Depends on the **structure** of the population. Town B looks twice as "sick" only because it is **older**.

Hypothetical data for adults aged 20+; edentulism = loss of all natural teeth.

Direct age standardisation

Apply the specific rates of **each** town to **the same standard population** (here: 7,000 people aged 20–64 and 3,000 aged 65+).

Age	**Expected, town A**	**Expected, town B**
20–64	7,000 × 0.02 = 140	7,000 × 0.02 = 140
65+	3,000 × 0.20 = 600	3,000 × 0.20 = 600
total	740 / 10,000 = **7.4%**	740 / 10,000 = **7.4%**

With the chosen age standard both towns have the same age-standardised prevalence — 7.4%.

A standardised measure is a **notional** number — valid for comparison only with the **same** standard, age groups and case definition. Crude numbers describe the real burden and are used for planning (how many dentures are needed).

Standardisation changes the weights, not age-specific frequencies

Continue the edentulism example: 2% at ages 20–64 and 20% at 65+ in both towns.

Population	Share aged 20–64	Share aged 65+	Weighted prevalence
Actual town A	80%	20%	$0.80 \times 2\% + 0.20 \times 20\%$ $= \mathbf{5.6\%}$
Actual town B	50%	50%	$0.50 \times 2\% + 0.50 \times 20\%$ $= \mathbf{11.0\%}$
Common standard	70%	30%	$0.70 \times 2\% + 0.30 \times 20\%$ $= \mathbf{7.4\%}$ for either town

Use the same age composition to compare frequencies. Use the actual population to calculate real service needs.

The mean (2% + 20%) / 2 = 11% assumes a 50:50 standard, not the chosen 70:30 standard.

Teaching calculations from the preceding two slides; direct age standardisation

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Data sources

Source	Examples (Bulgaria and international)	Strengths	Limitations
demographic and health statistics	National Statistical Institute (population, deaths by cause); National Centre of Public Health and Analyses — annual health statistics	full coverage, continuity	registered events only; few dental data
registries	National Cancer Registry; NCPHA cancer bulletins; oncology database in the National Health Information System (being built)	detailed follow-up of cases	completeness and timeliness must be assessed
administrative data	National Health Insurance Fund: reported dental activities under the National Framework Contract	large volume, cheap	show **services**, not disease
surveys	oral health survey following the WHO methodology; European Health Interview Survey (EHIS, NSI)	estimate of population prevalence, social factors	expensive, periodic; non-response; self-report is imprecise
model estimates	GBD (IHME), WHO country profiles	comparable across countries	**estimates**, not direct measurements

1200 visits do not mean 1200 people with disease

Hypothetical annual practice report

The practice records **1200 visits by 400 different patients**.
100 of those patients receive treatment for untreated caries. The area has 5000 residents.

Calculation	Result	Meaning
1200 / 400	**3 visits**	Mean per patient served
100 / 400	**25%**	Share treated for caries among practice attenders
100 / 5000	**2%**	Recorded people treated for caries relative to residents; not population prevalence

Estimating prevalence among all 5000 residents requires information on the health of non-attenders too.

Hypothetical administrative data; visits, unique patients and disease are different units

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In-class exercises

1 · Prevalence

500 people aged 35–44 are examined; 95 have at least one pocket ≥ 6 mm. What is the prevalence? Is it point or period prevalence?

2 · Cumulative incidence

200 caries-free six-year-olds are followed for 3 years with no losses; 50 develop caries. Calculate the risk.

3 · Person-time

Four people are followed for 3, 5, 2 and 4 years respectively; the second and fourth develop periodontitis at the end of their follow-up. Calculate the IR per 100 PY.

4 · Mortality and case-fatality

Population 400,000; 48 new cases of oral cancer in a year; followed for 12 months after diagnosis, 12 of them died of the cancer; 20 people in total died of this cause during the year. Calculate incidence, cause-specific mortality and case-fatality.

5 · $P \approx \mathrm{IR} \times D$

Incidence 0.5 per 1000 PY, mean duration 10 years. What is the expected prevalence?

6 · Crude rate

Aged 20–64: 6,000 people, 1% edentulous; 65+: 4,000 people, 25% edentulous. What is the crude prevalence?

Answers

No.	Solution	Answer
1	95 / 500; on the day of examination	**19%** point prevalence of the finding "pocket ≥ 6 mm"
2	50 / 200 over 3 years	**25%** over 3 years
3	$2 / (3 + 5 + 2 + 4) = 2 / 14$	**14.3 per 100 PY**
4	48 / 400,000; 20 / 400,000; 12 / 48	**12** and **5 per 100,000**; one-year case-fatality **25%**
5	$0.0005 \times 10 = 0.005$	**5 per 1000**
6	(60 + 1,000) / 10,000	**10.6%**

Before you divide, ask: who is at risk, for how long were they observed, and are we counting new or all cases?

Summary

Concepts

1. Epidemiology studies **frequency, distribution and determinants** in populations and applies them to control.
2. State the **case definition**, population, unit and time before calculating a measure.
3. The triad and the causal pie explain the multifactorial origin of caries and periodontitis.
4. The natural history defines the levels of prevention.

Measures

1. **Prevalence** — all cases; **incidence** — new cases among people at risk.
2. Cumulative incidence is a **risk**; the rate uses **person-time**.
3. $P \approx \mathrm{IR} \times D$.
4. **Mortality** — relative to the population; **case-fatality** — relative to the cases.
5. When populations differ in age structure, compare **specific** or **standardised** measures.

Next class: **Epidemiology of dental diseases** — oral health indices, WHO survey methodology, burden worldwide and in Bulgaria.

Thank you for your attention!